

Dr. Semmelweis - Savior of Mothers

DATA SCIENCE REPORT



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Dr. Semmelweis and Childbed Fever: A Data Analysis Report

In the mid-19th century, Dr. Ignaz Philipp Semmelweis, a Hungarian physician who worked at the General Hospital in Vienna, made a groundbreaking observation that would later revolutionize modern medicine. At a time when the causes of disease were still largely unknown and were often attributed to vague notions such as “bad air,” Semmelweis noticed the alarmingly high mortality rate among women in the maternity wards. His research led him to propose a simple yet revolutionary idea: that proper hygiene, particularly handwashing, could drastically reduce the number of deaths from puerperal fever. Although initially dismissed by many of his contemporaries, his findings would eventually become a cornerstone of antiseptic medical practice.

This report examines Semmelweis’s work from the perspective of modern data science to uncover the patterns and evidence underlying his conclusions. By analyzing historical data, we are trying not only to understand what he discovered, but also to what extent the data supports his claims. This approach illustrates just how effective data-driven thinking can be when it comes to challenging established beliefs and improving outcomes in practice. Ultimately, the goal is to show how even simple observations, when backed by data, can lead to life-saving innovations.

Source Data Files (CSV Format)

Annual deaths by clinics: [github.com](#)

Monthly deaths: [github.com](#)

Imported Libraries:

```
import pandas as pd
import numpy as np
import plotly.express as px
import matplotlib.pyplot as plt
import matplotlib.dates as mdates
from scipy.stats import gaussian_kde
from pandas.plotting import register_matplotlib_converters
import plotly.graph_objects as go
from plotly.subplots import make_subplots
```

Display Formatting for Financial Data

To improve readability and ensure consistent formatting, we are going to configure pandas to display floating-point numbers and values in a more appropriate way.

```
# Notebook Presentation:
pd.options.display.float_format = '{:,.2f}'.format
pd.set_option('display.max_columns', None)
pd.set_option('display.width', None)
```

To better understand the unusually high mortality rates observed by Ignaz Semmelweis, we will first examine—as he did—the historical records of births and maternal mortality in the 1840s. Using this data, we can move beyond mere speculation and look directly for patterns that might explain the tragic events in the maternity wards.

Acquiring the Source and Exploring Initial Dataset

We import two datasets containing annual and monthly records of births and deaths, which form the basis of our analysis.

```
df_yearly = pd.read_csv('csv_files/annual_deaths_by_clinic.csv')
df_monthly = pd.read_csv('csv_files/monthly_deaths.csv')
```

Converting Date Columns to Datetime Format

To ensure accurate time-based analysis and enable proper visualization, we convert date-related columns from strings into a standardized date and time format.

```
df_yearly["year"] = pd.to_datetime(df_yearly["year"], format="%Y")
df_monthly.date = pd.to_datetime(df_monthly.date)
```

Getting initial Insight

```
print("Yearly Data:")
print(df_yearly)
print("\nMonthly Data:")
print(df_monthly)
```

Yearly Data:

	year	births	deaths	clinic
0	1841-01-01	3036	237	clinic 1
1	1842-01-01	3287	518	clinic 1
2	1843-01-01	3060	274	clinic 1
3	1844-01-01	3157	260	clinic 1
4	1845-01-01	3492	241	clinic 1
5	1846-01-01	4010	459	clinic 1
6	1841-01-01	2442	86	clinic 2
7	1842-01-01	2659	202	clinic 2
8	1843-01-01	2739	164	clinic 2
9	1844-01-01	2956	68	clinic 2
10	1845-01-01	3241	66	clinic 2
11	1846-01-01	3754	105	clinic 2

Monthly Data:

	date	births	deaths
0	1841-01-01	254	37
1	1841-02-01	239	18
2	1841-03-01	277	12
3	1841-04-01	255	4
4	1841-05-01	255	2
..	...	...	...
93	1848-11-01	310	9
94	1848-12-01	373	5
95	1849-01-01	403	9
96	1849-02-01	389	12
97	1849-03-01	406	20

[98 rows x 3 columns]

As illustrated above, the two datasets cover the years from 1841 to 1849.

The tables show the records of total births and maternal deaths during childbirth, allowing us to analyze mortality patterns over time.

Furthermore, the yearly data also provides the number of births and deaths down by two clinics at Vienna Hospital.

Checking Missing and Duplicate Values

Before we proceed with the analysis, it is essential to verify the integrity of the datasets.

By checking for missing and duplicate values, we ensure that our results are reliable and not skewed by incomplete or duplicate records.

```
print("\nIs there any empty or duplicated values on Yearly data:")
print(f"Missing Values: {df_yearly.isna().values.any()}")
print(f"Duplicated Values: {df_yearly.duplicated().any()}")

print("\nIs there any empty or duplicated values on Monthly data:")
print(f"Missing Values: {df_monthly.isna().values.any()}")
print(f"Duplicated Values: {df_monthly.duplicated().any()}")
```

Is there any empty or duplicated values on Yearly data:

Missing Values: False

Duplicated Values: False

Is there any empty or duplicated values on Monthly data:

Missing Values: False

Duplicated Values: False

As reflected in the results, the datasets contain no missing or duplicate entries; this makes it possible to begin the analysis directly without further investigation.

Feature Engineering: Creating a Mortality Rate Column

To gain a deeper understanding of the data, we are going to create a new columns that represent the mortality rate by calculating the ratio of the number of deaths.

This transformation allows us to look beyond raw data and compare results more meaningfully across different time periods and clinics, revealing underlying patterns that might not be apparent from the raw data alone.

```
# Creating Death Percentages Data:
df_yearly['pct_deaths'] = df_yearly.deaths / df_yearly.births * 100
df_monthly['pct_deaths'] = df_monthly.deaths / df_monthly.births * 100
```

Average Maternal Mortality Rate in the 1840s

To better understand the scale of the situation, we calculate the overall mortality rate by comparing the total number of maternal deaths with the total number of births over the entire decade.

This provides a single, aggregated value that reflects the overall risk of giving birth at Vienna General Hospital during this period.

```
mortality_rate = df_yearly.deaths.sum() / df_yearly.births.sum() * 100

print(f'The mortality rate among women who died in childbirth was:
{mortality_rate:.3}%')
```

The mortality rate among women who died in childbirth was 7.08%.

The calculated mortality rate of 7.08% highlights the alarmingly high risk associated with childbirth in the 1840s—meaning that about 7 out of every 100 women did not survive. To put this into perspective, maternal mortality rates in Europe today stand at around 0.007% (7 per 100,000 births).

This stark contrast illustrates just how dangerous childbirth used to be and highlights the absence of basic medical procedures that we now consider indispensable.

Statistical Overview of Childbirth and Maternal Mortality

To better understand the structure and distribution of the data, we are going to examine key descriptive statistics with using `.describe()` function for both the annual and monthly datasets.

This overview provides a summary of the central tendencies, dispersion, and outliers, and helps us identify patterns and potential anomalies before conducting a more in-depth analysis.

```
print("\nDescriptive Analysis of the Dataframes:")
print(df_yearly.describe())
print(df_monthly.describe())
```

Yearly Data:

	year	births	deaths	pct_deaths
count	12	12.00	12.00	12.00
mean	1843-07-02	3,152.75	223.33	6.95
min	1841-01-01	2,442.00	66.00	2.04
25%	1842-01-01	2,901.75	100.25	3.34
50%	1843-07-02	3,108.50	219.50	7.25
75%	1845-01-01	3,338.25	263.50	8.42
max	1846-01-01	4,010.00	518.00	15.76
std	NaN	449.08	145.38	4.05

Monthly Data:

	date	births	deaths	pct_deaths
count	98	98.00	98.00	98.00
mean	1845-02-11	267.00	22.47	8.62
min	1841-01-01	190.00	0.00	0.00
25%	1843-02-08	242.50	8.00	2.91
50%	1845-02-15	264.00	16.50	6.22
75%	1847-02-22	292.75	36.75	13.32
max	1849-03-01	406.00	75.00	31.38
std	NaN	41.77	18.14	7.18

The annual data show an average of approximately 3,150 births and 223 maternal deaths, corresponding to an average mortality rate of about 7%; however, the wide range (from 2.04% to 15.76%) indicates significant fluctuations that are more likely due to external factors than to natural variations.

This pattern is even more evident in the monthly data, where mortality rates exceed 30% in some cases, strongly suggesting systemic problems within the hospital environment—particularly poor hygiene practices.

Despite relatively stable birth rates, the disproportionately large fluctuations in the number of deaths suggest that mortality was determined more by the quality of care than by patient volume, which reinforces the assumption of a preventable cause.

Analysis of total number of Births and Death in childbirth

Before major medical procedures were introduced, it is important to understand the conditions that prevailed in the maternity wards of Vienna General Hospital between 1841 and 1847. By examining both the number of births and the number of maternal deaths over time, we can identify patterns that may explain the persistently high mortality rates.

This period reflects the unaltered, original state of clinical practice and offers insights into the environment that Ignaz Semmelweis studied in depth before reaching his groundbreaking conclusions.

Visualizing Data with a Dual-Axis Line Chart

To better compare trends in birth and death rates over time, we create a two-axis line graph that allows both variables to be displayed simultaneously, despite their different scales. This approach makes it easier to observe how fluctuations in the number of deaths relate to the total number of births over the same period.

code:

```
fig = make_subplots(specs=[[{"secondary_y": True}]])

color_births = "#2E86C1"
color_deaths = "#E23E57"

# Plotting Birth Line
fig.add_trace(
    go.Scatter(
        x=df_monthly['date'],
        y=df_monthly['births'],
        name="Births",
        mode="lines",
        line=dict(color=color_births, width=3),
        hovertemplate="Births: %{y}<extra></extra>"
    ),
    secondary_y=False,
)

# Plotting Death Line
fig.add_trace(
    go.Scatter(
        x=df_monthly['date'],
        y=df_monthly['deaths'],
        name="Deaths",
        mode="lines",
```



```

        line=dict(color=color_deaths, width=3, dash="dot"),
        hovertemplate="Deaths: %{y}<extra></extra>"
    ),
    secondary_y=True,
)

# Designing the Layout
fig.update_layout(
    title=dict(
        text="<b>Total Number of Births and Deaths</b> (1841 - 1847)",
        font=dict(size=22, family="Arial, sans-serif", color="#52057B"), x=0.5),
    plot_bgcolor="white", paper_bgcolor="white",
    hovermode="x unified",
    hoverlabel=dict(bgcolor="white", font_size=14, font_family="Arial"),
    legend=dict(orientation="h", yanchor="bottom", y=1.05, xanchor="right", x=1),
    margin=dict(l=60, r=60, t=100, b=60)
)

# Refining X-axis
fig.update_xaxes(
    title_text="",
    showgrid=True, gridwidth=1, gridcolor="rgba(230, 230, 230, 0.8)",
    dtick="M12", tickformat="%Y", # ticks for years at the x-axis
    tickfont=dict(color="#555555"),
    ticks="outside",
    ticklen=8, # Years should have longer ticks here
    minor=dict( # Ticks for months
        dtick="M1", # for every month
        ticklen=4, # shorter ticks this time
        tickcolor="#cccccc", # little open black
        ticks="outside"
    ),
    hoverformat="%B %Y",
    range=[df_monthly['date'].min(), pd.to_datetime("1847-04-01")]
)

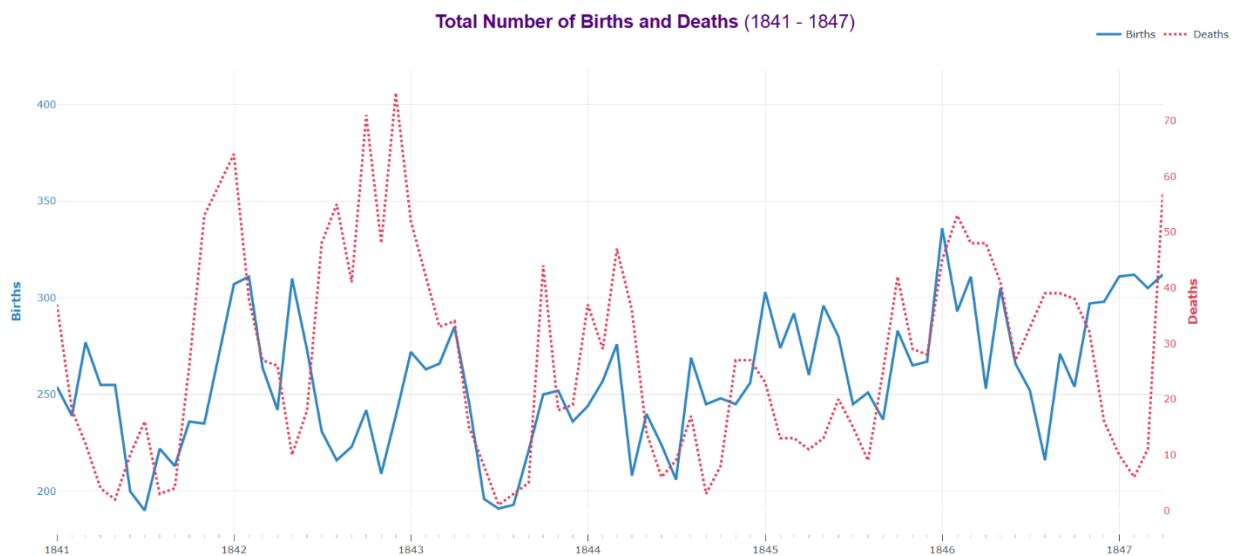
# Refining y-axis (for births)
fig.update_yaxes(
    title_text="<b>Births</b>",
    title_font=dict(color=color_births),
    tickfont=dict(color=color_births),
    showgrid=True, gridwidth=1,
    gridcolor="rgba(230, 230, 230, 0.8)",
    secondary_y=False
)

# Refining y-axis (for deaths)
fig.update_yaxes(
    title_text="<b>Deaths</b>", title_font=dict(color=color_deaths),
    tickfont=dict(color=color_deaths),
    showgrid=False,
    secondary_y=True
)

```

```
#fig.write_html("interactive_graphs/semmelweis/total_births_and_deaths_until_1847.html")  
  
fig.show()
```

Graph:



Interactive graph link: [Total Number of Births and Deaths](#)

The graph clearly shows that, maternal mortality has fluctuated sharply over time, with significant peaks—particularly around 1842–1843—suggesting that mortality was not solely dependent on the number of patients.

These irregular spikes are more likely to indicate underlying systemic factors, such as inconsistent medical practices or environmental conditions, rather than natural fluctuations; in the past, such patterns have been linked to contamination risks in clinical settings, particularly where hygiene protocols were lacking.

The interactive features of the graph, such as hovering and applying filters, also highlight how sudden and concentrated these spikes are, supporting the assumption that specific events or practices—rather than gradual trends—were responsible for this dangerous rise in mortality.

Analysis of Different Clinics at the Hospital

To gain a deeper insight into conditions at Vienna General Hospital, we are going to compare data from its two maternity wards, each of which operates under different staffing conditions.

Clinic 1, which is staffed by doctors and medical students, and Clinic 2, which is run by midwives, provide a natural comparison for examining how medical practices may have influenced the outcomes.

By analyzing births and deaths at these clinics, we aim to identify differences in workload, changes in patient trends over time, and fluctuations in mortality rates that may point to underlying causes.

Visualizing Mortality Rates by Clinic

To enable a clear comparison between the two hospitals, we are going to use an interactive line chart to visualize how mortality rates evolve over time for each hospital, leveraging Plotly features such as markers and hover data to provide detailed and user-friendly insights into the underlying data.

```
df_yearly['proportion_deaths'] = df_yearly['deaths'] / df_yearly['births']

fig = px.line(
    df_yearly,
    x='year',
    y='proportion_deaths',
    color='clinic',
    markers=True,
    hover_data={
        'proportion_deaths': ':.1%',
        'births': True,
        'deaths': True
    },
    labels={
        'year': 'Year',
        'proportion_deaths': 'Mortality Rate',
        'clinic': 'Clinic',
        'births': 'Total Births',
        'deaths': 'Total Deaths'
    },
    template='plotly_white',
    color_discrete_sequence=['#00B8A9', '#F6416C']
)
```

```

# Designing the layout
fig.update_layout(
    title=dict(
        text='<b>Yearly Mortality Rate by Clinic</b><br><sup>Proportion of deaths</sup> per birth at Vienna General Hospital (1841-1846)</sup>',
        font=dict(size=23, family="Arial, sans-serif", color="#406AAF")
    ),
    font_family="Arial",
    hovermode="x unified",
    legend=dict(orientation="h", yanchor="bottom", y=1.02, xanchor="right", x=1)
)

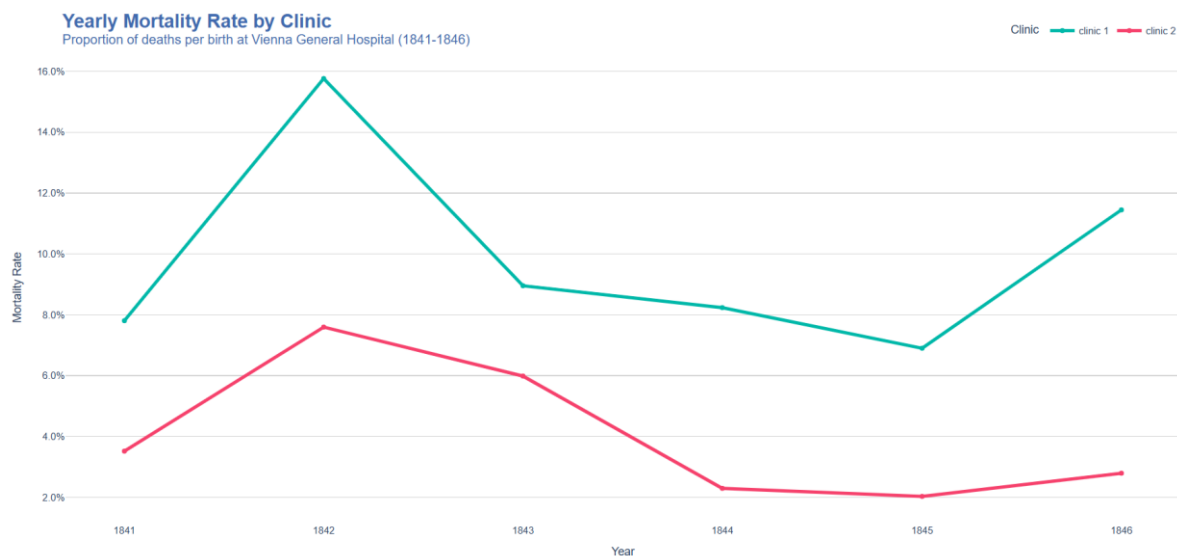
fig.update_yaxes(tickformat=".1%", showgrid=True, gridcolor='LightGray')
fig.update_xaxes(showgrid=False)
fig.update_traces(line=dict(width=4))

# fig.write_html("interactive_graphs/semmelweis/clinic_comparison.html")

fig.show()

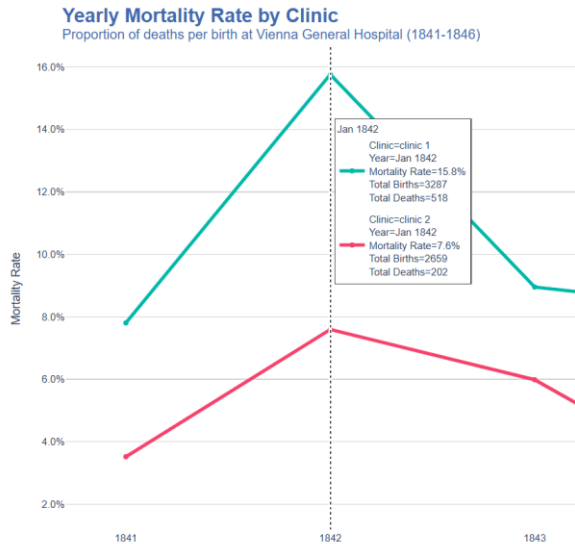
```

Graph:



Interactive graph link: [Yearly Mortality Rate by Clinic](#)

The graph shows a consistently higher mortality rate at Clinic 1 (which is staffed by doctors and medical students) compared to Clinic 2 (which is run by midwives), with the difference being particularly pronounced around the year 1842: While Clinic 1



reached a peak of nearly 16%, Clinic 2 remained below 8%.

When we hover over the data points, we can see that Clinic 1 often records slightly more births (e.g., over 3,000 per year) and significantly higher maternal deaths (over 500 in peak years), suggesting that the difference is substantial not only in relative terms but also in absolute numbers.

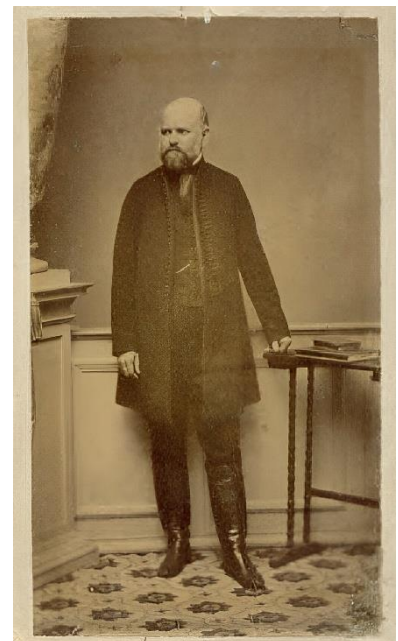
This persistent inconsistency shed light on systematic differences in clinical practice rather than on patient volume, prompting

Dr. Ignaz Semmelweis to investigate the different procedures used in Clinic 1.

Semmelweis' Breakthrough: Uncovering the Truth Behind Childbed Fever

At first, Dr. Semmelweis taught that the position of the women in labor was the problem. In Clinic 2 (the midwives' clinic), women gave birth in the lateral position. In the doctors' clinic, women gave birth on their backs. So Dr. Semmelweis had the women in the doctors' clinic give birth in the lateral position. However, this had no effect on the mortality rate.

Next, Dr. Semmelweis noticed that whenever someone died on the clinic 1, a priest would walk through, passing by the women's beds and ringing a bell as he went. Perhaps the priest and the ringing of the bell frightened the women



so much after childbirth that they developed a fever, fell ill, and died. Dr. Semmelweis had the priest change his route and stop ringing the bell. This, too, had no effect.

At that point, Dr. Semmelweis was so frustrated that he took a vacation in Venice. Perhaps a short break would clear his head. When Semmelweis returned from his vacation, he learned that one of his colleagues, a pathologist, had fallen ill and died. His friend had pricked his finger during the autopsy of a woman who had died of childbed (puerperal) fever and had subsequently become seriously ill and died himself. (At that time, people were not yet fully aware of viruses and bacterial diseases or their contagious nature.)

When Semmelweis examined the pathologist's symptoms, he realized that the man had died of the same disease as the women he had performed autopsies on. This was his breakthrough: anyone could get sick from childbed fever, not just women giving birth!

This led to Semmelweis's new theory. It was possible that small fragments or particles from a corpse had come into contact with the hands of the doctors and medical students while they were dissecting the bodies during an autopsy. And when the doctors in Clinic 1 delivered the babies, these particles entered the bodies of the women in labor, who then contracted the disease and died.

The Effect of Handwashing

In June 1847, Ignaz Semmelweis introduced a new regulation at the Vienna General Hospital requiring medical staff to clean their hands and instruments with a **chlorine solution**. Although the scientific understanding of bacteria and germs was not yet fully developed, Semmelweis observed that this approach significantly reduced the unpleasant odors associated with autopsies and suspected a connection to the patients' course of illness.

This decision, which was based more on observations than on theoretical considerations, would later prove to be one of the most significant breakthroughs in the history of medical hygiene.

Splitting the Dataset Before and After Intervention

To assess the impact of this measure, we divide the dataset into two time periods: before and after the introduction of handwashing. This method, commonly used in data analysis, allows us to conduct a comparative analysis and measure how a particular measure affects the results over different time periods.

```
new_policy_date = pd.Timestamp('1847-06-01')

before_washing_hand = df_monthly.query("date < @new_policy_date")
after_washing_hand = df_monthly.query("date >= @new_policy_date")

before_avg_rate =
    before_washing_hand.deaths.sum() / before_washing_hand.births.sum() * 100
after_avg_rate =
    after_washing_hand.deaths.sum() / after_washing_hand.births.sum() * 100

print(f'Average death rate before 1847 was {before_avg_rate:.4}%')
print(f'Average death rate After 1847 is {after_avg_rate:.3}%')
```

```
Average death rate before 1847 was 10.53%
Average death rate After 1847 is 2.15%
```

The results show a dramatic decline in the mortality rate from 10.53% before the intervention to just 2.15% afterward, indicating a significant improvement in treatment outcomes.

Such a sharp decline strongly suggests a causal link between handwashing and a lower mortality rate and provides compelling evidence that simple hygiene measures played a crucial role in saving lives—even before the scientific reasons behind this were fully understood.

Analysis of the Impacts of Antiseptic Practice

In this section, we are going to examine the effects of the hygiene measures introduced by Ignaz Semmelweis, which are now regarded as an early form of antiseptics. By visualizing the data surrounding this turning point, we aim to illustrate how mortality trends have changed since the introduction of systematic handwashing.

Smoothing Trends with a Rolling Average

To better understand the underlying trend in mortality rates, we use a 6-month moving average to smooth out short-term fluctuations and highlight longer-term patterns.

To perform this time-based operation correctly, we set the date column as the index to ensure that the data is treated as a continuous time series and prevents the temporal structure from being lost during the calculation.

```
roll_df = before_washing_hand.set_index('date')
roll_df = roll_df.rolling(window=6).mean()
```

Plotting the graph with turning point:

Here, we are plotting a line graph that distinguishes between mortality trends before and after the intervention, allowing us to visually compare the two time periods in a single, clear chart.

By highlighting this turning point, the visualization illustrates how mortality patterns evolve over time, making it easier to identify structural changes.

code:

```
fig, ax = plt.subplots(figsize=(16, 8), dpi=300)

# Plotting the data before 1847
ax.plot(roll_df.index,
        roll_df.pct_deaths,
        color='#D14D72',
        linewidth=2.5,
        label='Before Handwashing (6m Moving Average)')

# Plotting the data after 1847
ax.plot(after_washing_hand.date,
        after_washing_hand.pct_deaths,
        color='#FF9A00',
        linewidth=2.5,
        marker='o', markersize=6,
        markerfacecolor='white', markeredgewidth=1.5,
        label='After Handwashing')

# The turning point:
ax.axvline(x=after_washing_hand.date.min(), color='#b0b0b0', linestyle='--',
```



```

linewidth=1.5, zorder=0)

# Designing the Layout
ax.set_title('Percentage of Monthly Maternal Deaths over Time',
             fontsize=16, fontweight='bold', color='#102C57', pad=20, loc='left')
ax.set_ylabel('Percentage of Deaths (%)', fontsize=13, color=text_color)
ax.set_xlabel('')
ax.set_xlim([df_monthly.date.min(), df_monthly.date.max()])

# Ticks at x-axis
ax.tick_params(axis='both', which='major', labels=12, colors=text_color)
plt.xticks(rotation=0)

ax.grid(axis='y', color='#e0e0e0', linestyle='--', linewidth=0.8)
ax.grid(axis='x', visible=False)

# A clear open graph
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
ax.spines['left'].set_visible(False)

ax.legend(frameon=False, fontsize=12, loc='upper right', labelcolor=text_color)
plt.tight_layout()

plt.show()

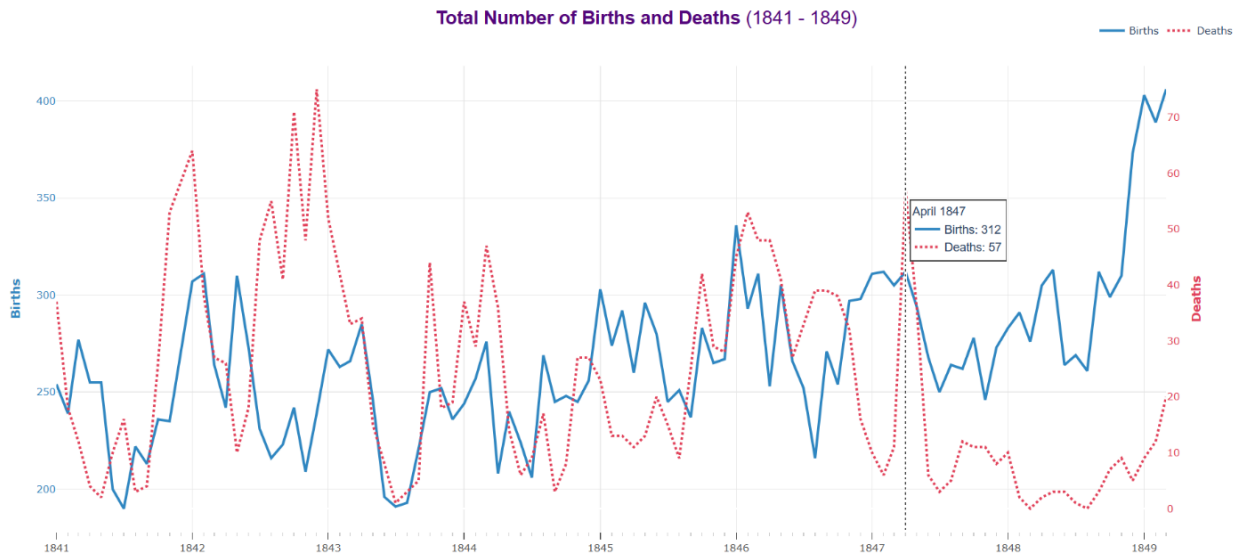
```

Graph:



The Graph shows the total number of Births and Maternal Deaths:

We have created this graph before.



Interactive Graph link:

The graph shows a dramatic and immediate decline in maternal mortality following the introduction of handwashing with a chlorine solution by Ignaz Semmelweis in 1847: Mortality rates fell from fluctuating levels—often exceeding 10–15%—to consistently below 5%, and in some months even approached zero.

This marked shift is particularly noteworthy because it is occurring despite relatively stable birth rates (as shown in the second graph), which strongly suggests that the decline in deaths was not due to a smaller patient population, but rather to a fundamental improvement in clinical conditions.

From today's perspective, such a sudden and sustained change is characteristic of an effective intervention; this supports the conclusion that antiseptic practices directly reduced infection-related mortality and represents one of the earliest clear examples of a data-driven medical breakthrough.

Strengthening the Argument: Statistical Evidence and Distribution Analysis

To better illustrate the effectiveness of handwashing, we are going to go beyond simple observations and rely on statistical analyses combined with visual representations.

By analyzing how the distribution of mortality rates changes over time, we can develop a more compelling, data-driven argument that highlights not only trends but also measurable differences in outcomes.

Comparing Average Monthly Mortality Rates

In this section, we compare the average monthly mortality rates before and after the introduction of handwashing to quantify its impact.

By calculating both the absolute difference and the relative improvement, we aim to clearly demonstrate how significantly the risk of dying during childbirth has improved and how this compares to the general trends of the 1840s.

```
mean_diff = before_avg_rate - after_avg_rate
improvement_rate = before_avg_rate / after_avg_rate

print(f"Hemmelweis' introduction of antiseptics reduced the mortality rate by
{mean_diff:.3}%!")
print(f'This is a {improvement_rate:.2}x improvement!')
```

```
Semmelweis' introduction of antiseptics reduced the mortality rate by 8.37%!
This is a 4.9x improvement!
```

The results show a dramatic decline in mortality: average mortality rates fell from around 10.5% to just over 2%, representing an absolute decrease of 8.37 percentage points. This represents a nearly fivefold improvement, meaning that the likelihood of dying during childbirth has decreased by about five times among women since the introduction of handwashing.

Such a clear and sustained change provides strong statistical evidence that the intervention was highly effective, supporting the conclusion that hygiene measures played a crucial role in reducing maternal mortality.

Comparative Statistical Analysis of Mortality Distributions

While averages provide a good indication of changes, they do not fully reflect how the overall distribution of mortality rates has evolved over time. To account for this, we use box plots to visualize key statistical measures such as the median, quartiles, and outliers, which gives us a more comprehensive picture of variability and consistency.

This approach allows us not only to compare the overall trend, but also to assess how much mortality rates fluctuated before and after the intervention and how unpredictable they were. In this way, we can determine whether the improvements have been consistent over several months or are due only to a few outliers.

Forming the DataFrame:

To enable this comparison, we create a new categorical variable that classifies each observation as occurring either before or after the introduction of handwashing. This is achieved using the NumPy function `.where()`, which efficiently assigns values based on a condition making it a powerful tool for conditional data transformation.

In addition, we are adding extra fields to enhance interactivity, so that detailed information—such as specific dates, births, and deaths—is displayed when you hover over individual data points.

```
df_monthly["handwashing"] =  
    np.where(df_monthly.date >= new_policy_date, 'After', 'Before')  
  
df_monthly['date_str'] = df_monthly['date'].dt.strftime('%Y-%m-%d')  
  
display_data=["date_str", "pct_deaths", "births", "deaths"]
```

Creating Box Plot

Here, we create a box plot to compare the distribution of monthly mortality rates between the two time periods, using colors and individual data points to enhance the clarity of the analysis.

This visualization makes it easy to identify shifts in the median values, a reduction in variability, and changes in the outliers following the intervention.

```
fig = px.box(
    df_monthly,
    x="handwashing",
    y="pct_deaths",
    color="handwashing",
    title="Impact of Handwashing on Monthly Death Rate",
    points="all",
    template="plotly_white",
    color_discrete_map={"Before": "#DA0037", "After": "#6FAF4F"}, # Hover-template
    custom_data=display_data
)

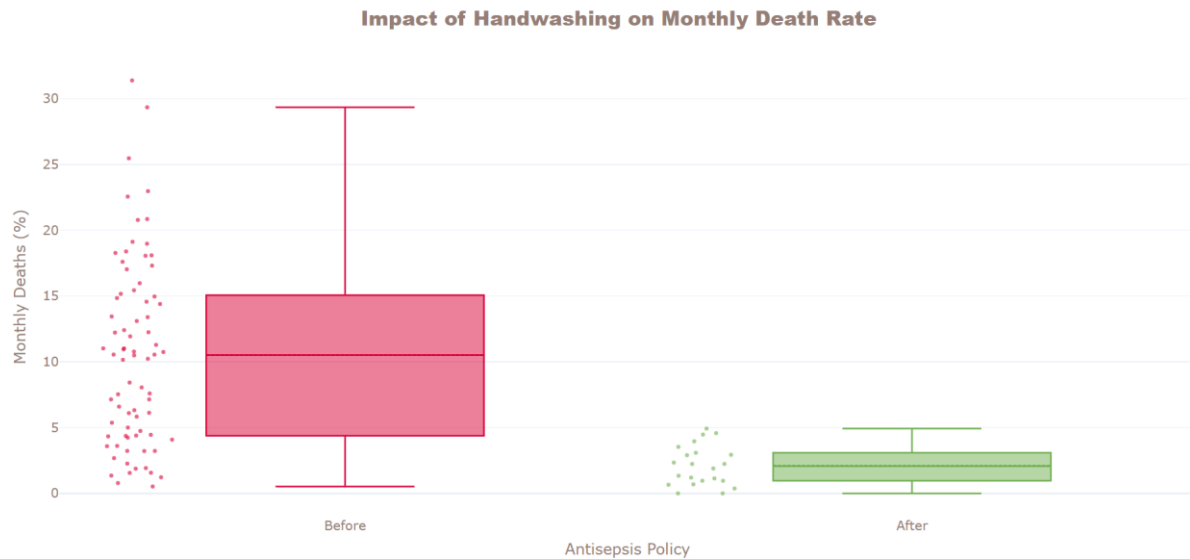
# Refining the Layout
fig.update_layout(
    title_font_size=25,
    title_font_family="Arial Black",
    title_x=0.5,
    font=dict(size=16, color="#967E76"),
    showlegend=False,
    margin=dict(t=80, b=40, l=60, r=40),
    yaxis_title="Monthly Deaths (%)",
    xaxis_title="Antisepsis Policy",
    hoverlabel=dict(bgcolor="white", font_size=15, font_family="Arial, sans-serif")
)

# Customizing the hover template
fig.update_traces(
    marker=dict(size=5, opacity=0.7, line=dict(width=0)),
    line=dict(width=2),
    boxmean=True,
    hovertemplate=(
        "<b>Death Rate:</b> %{customdata[1]:.1f}%<br>"
        "<b>Date:</b> %{customdata[0]}<br>"
        "<b>Total Births:</b> %{customdata[2]}<br>"
        "<b>Total Deaths:</b> %{customdata[3]}<br>"
        "<extra></extra>"
    )
)

#
fig.write_html("interactive_graphs/semmelweis/before_and_after_handwashing.html")

fig.show()
```

Box plot:



Interactive graph link: [Impact of Handwashing on Monthly Death Rate](#)

The box plot illustrates a dramatic shift not only in the average mortality rate but also in the overall distribution: the median fell from about 10.5% to approximately 2%, and the upper quartile from about 15% to just over 3%, suggesting that even the “typically worst months” became significantly safer following the measure.

Another striking observation is the disappearance of extreme outliers: Before the handwashing regulations were introduced, the mortality rate could rise to over 30% (as was observed in some months), whereas since the regulations were introduced, the figures have remained just under 5%, suggesting that catastrophic outbreaks have been effectively prevented.

From a statistical perspective, this decline in both the central tendency and the variability strongly supports the conclusion that Ignaz Semmelweis’s intervention not only improved outcomes on average but also fundamentally stabilized the system by eliminating the cause of the massive spikes in mortality.

Visualizing Distribution Shapes with Histograms

While box plots summarize key statistical measures, they do not provide a complete picture of how frequently different mortality rates occur over time. To gain a deeper understanding of the data distribution, we are going to use histograms to illustrate how monthly death rates are distributed before and after the measure.

In this way, we can observe not only shifts in the average values, but also how the overall distribution, frequency, and range of the results change following the introduction of antiseptic measures.

Creating Histogram:

We use Plotly to create overlapping histograms to compare the distributions of monthly mortality rates for the two time periods, distinguishing between “before” and “after” handwashing with different colors. To ensure a fair comparison despite differences in time periods, we normalize the data using percentage frequencies, while transparency (opacity) allows us to view both distributions simultaneously.

In addition, we improve the presentation by adjusting the number of bins to clearly illustrate the range of the results, and by including a marginal box plot that provides a concise summary of the key statistical measures alongside the histogram.

```
fig = px.histogram(  
    df_monthly,  
    x="pct_deaths",  
    color="handwashing",  
    opacity=0.75,  
    nbins=30, # How many box do we want here  
    barmode="overlay",  
    histnorm="percent",  
    marginal="box", # show statistical plots at the top  
    title="Distribution of Monthly Death Rates: Before vs. After Antisepsis Policy",  
    template="plotly_white",  
    color_discrete_map={"Before": "#DF2E38", "After": "#48A111"},  
)  
  
# Refining the Layout  
fig.update_layout(  
    title=dict(font=dict(size=22, family="Arial Black", color="#A47251"), x=0.05),  
    font=dict(size=14, color="#333333"),  
    xaxis_title="Monthly Death Rate (%)", yaxis_title="Frequency (% of Months)",  
    legend=dict(  

```

```

        title="<b>Antisepsis Policy</b>",
        orientation="h", xanchor="right", yanchor="bottom", # Horizontal Legend
        x=1, y=1.02,          # The Legend will be above the chart.
    ),
    margin=dict(t=120, b=50, l=60, r=40),
    hoverlabel=dict(font_size=14, font_family="Arial, sans-serif")
)
# Subtle gridlines:
fig.update_xaxes(showgrid=True, gridwidth=1, gridcolor='#F0F0F0', zeroline=False)
fig.update_yaxes(showgrid=True, gridwidth=1, gridcolor='#F0F0F0', zeroline=False)

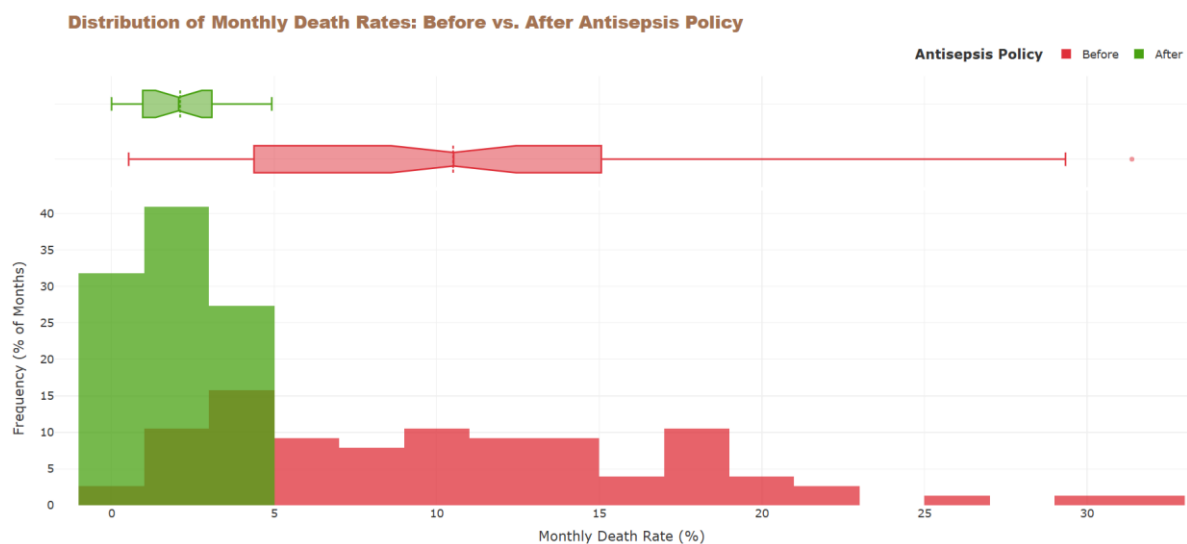
# Hover arrangements:
fig.update_traces(
    hovertemplate=(
        "<b>Death Rate Range:</b> %{x}%<br>"
        "<b>Frequency:</b> %{y:.1f}% of observed months<br>"
        "<extra></extra>"),
    selector=dict(type="histogram") # marginal box plot won't break at hovers
)

fig.update_traces(boxmean=True, marker=dict(opacity=0.5),
    selector=dict(type="box"))

fig.show()

```

Graph:



Interactive Graph Link: [Distribution of Monthly Death Rates](#)

The histogram clearly shows a complete shift in the distribution: Before the Handwashing policy was implemented, mortality rates were widely scattered between about 5% and over 20%, with isolated outliers above 30%, whereas after handwashing, the data are tightly clustered between 0% and 5%, with most values clustered around 1–3%. This compression of the distribution not only confirms a lower average value but also suggests that high-risk months have virtually disappeared.

This strongly suggests that Ignaz Semmelweis's antiseptic measures eliminated the actual cause of the fluctuations, likely the transmission of infections—and transformed childbirth from an unpredictable and dangerous event into a far more stable and controlled process.

Smoothing Distributions with Kernel Density Estimation (KDE)

A kernel density estimate (KDE) is a statistical method used to transform discrete data points into a smooth, continuous probability distribution, which allows us to better understand the underlying structure of the data.

Instead of grouping the values into classes, KDE applies a small, smooth curve (kernel) to each data point and combines them into a single, continuous curve that represents the probability of different mortality rates.

This approach provides a clearer and more nuanced picture of the distribution of mortality rates, thereby making it easier to compare how the overall picture has changed before and after the introduction of antiseptic measures.

Forming the DataFrame:

To plot the KDE curves, we generate a continuous range of values and apply a density estimation method using SciPy's *gaussian_kde()*, which models the distribution as a smooth probability function.

By analyzing these functions across a finely divided range, we convert discrete observational values into continuous curves, making it easier to compare patterns and identify shifts in the distribution between the two time periods.

One limitation of KDE's default settings is that the estimated curve can extend into unrealistic ranges (such as negative mortality rates), which are not meaningful in this

context. To address this, we apply constraints (such as clipping the range) to ensure the visualization remains within valid and interpretable boundaries.

```
# We use 500 points to make the curve perfectly smooth
x_range = np.linspace(0, 40, 500)

# Calculation of Kernel Density Estimates using SciPy
kde_before = gaussian_kde(before_washing_hand['pct_deaths'])
kde_after = gaussian_kde(after_washing_hand['pct_deaths'])

# Evaluating of the KDEs across the defined x_range
y_before = kde_before(x_range)
y_after = kde_after(x_range)
```

Creating KDE plot:

```
fig = go.Figure()

# Plotting the data before 1847
fig.add_trace(go.Scatter(
    x=x_range,
    y=y_before,
    fill='tozeroy', # Fill the bottom of the line
    mode='lines',
    line=dict(color='#EF553B', width=3),
    fillcolor='rgba(239, 85, 59, 0.4)',
    name='Before Antisepsis',
    hovertemplate="<b>Death Rate:</b> {x:.1f}%<br><b>Density:</b> {y:.4f}<extra></extra>"
))

# Plotting the data after 1847
fig.add_trace(go.Scatter(
    x=x_range,
    y=y_after,
    fill='tozeroy',
    mode='lines',
    line=dict(color='#00CC96', width=3),
    fillcolor='rgba(0, 204, 150, 0.4)',
    name='After Antisepsis',
    hovertemplate="<b>Death Rate:</b> {x:.1f}%<br><b>Density:</b> {y:.4f}<extra></extra>"
))

# Refining the layout
fig.update_layout(
```

```

title=dict(text="Estimated Distribution of Monthly Death Rates",
            font=dict(size=22, family="Arial Black", color="#A47251"), x=0.05),
template="plotly_white",
font=dict(size=14, color="#333333"),
xaxis_title="Monthly Death Rate (%)",
yaxis_title="Estimated Density",

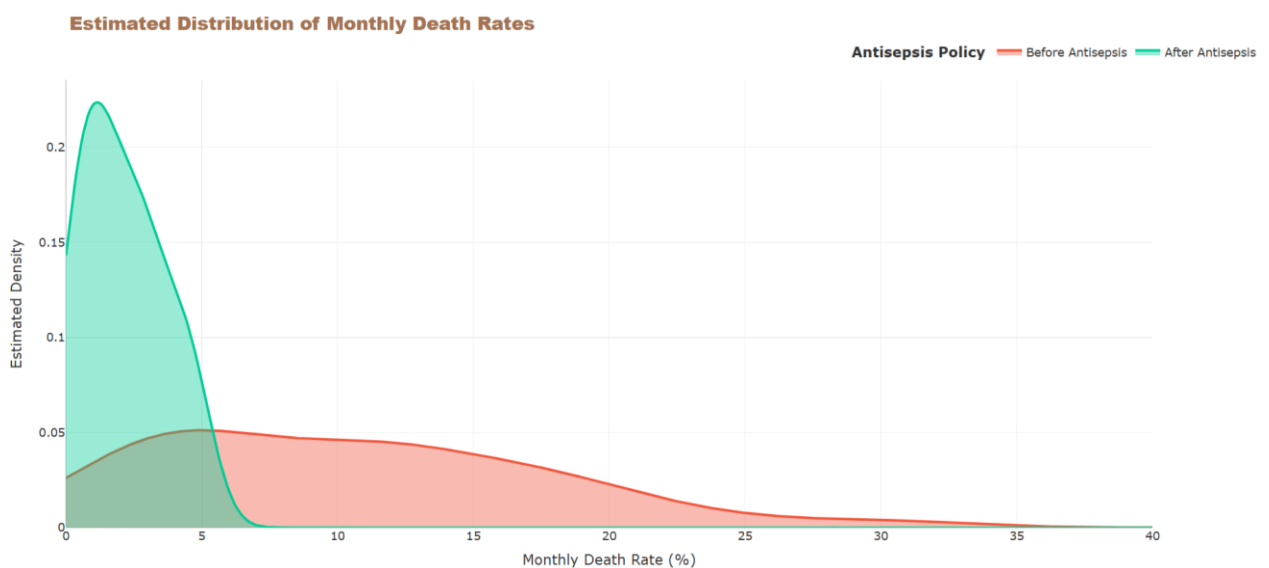
# Legend formatting and placement
legend=dict(
    title="<b>Antisepsis Policy</b>",
    orientation="h",
    yanchor="bottom", xanchor="right",
    y=1.03, x=1.1),
margin=dict(t=120, b=50, l=60, r=40),
hoverlabel=dict(bgcolor="white", font_size=14, font_family="Arial")
)

# Subtle gridlines
fig.update_xaxes(showgrid=True, gridwidth=1, gridcolor='#F0F0F0', zeroline=True)
fig.update_yaxes(showgrid=True, gridwidth=1, gridcolor='#F0F0F0', zeroline=True)

fig.show()

```

Graph:



Interactive graph: [Estimated Distribution of Monthly Death Rates](#)

The KDE curve confirms the same finding as the histogram, but presents it in a smoother, more continuous form, making the shift in mortality distributions before and after the introduction of antisepsis even more visually intuitive.

While histograms depict frequency based on classes, the KDE highlights the overall shape and probability density, clearly showing how probabilities associated with high mortality rates disappear almost entirely after the policy expires, while those associated with low mortality rates dominate.

As a result, KDE is particularly well-suited for identifying underlying patterns and shifts in the distribution, confirming that the decline in mortality was not only significant in magnitude but also consistent and systemic, rather than being driven by a few outliers.

Statistical Validation with Hypothesis Testing (T-Test)

To thoroughly assess whether the observed decline in mortality rates is meaningful, we use an independent two-sample t-test, a statistical method that compares the means of two groups and determines whether the difference between them is statistically significant.

In this context, we are going to examine whether the difference in the average monthly mortality rate before and after the introduction of antisepsis is greater than would be expected due to random fluctuations alone.

By analyzing the resulting p-value using a strict significance threshold of 1%, we can determine with a high degree of certainty whether handwashing (antisepsis) had a real, measurable impact on patient outcomes.

Computing the Test Statistics

Using a statistical library, we perform a two-sample t-test that calculates both the t-statistic (which measures the magnitude of the difference relative to the variability of the data) and the p-value (which quantifies the probability that such a difference could occur by chance).

This approach allows us to move beyond visual patterns and descriptive statistics and proceed to formal statistical inference.

```
import scipy.stats as stats

t_stat, p_value = stats.ttest_ind(a=before_washing_hand.pct_deaths,
b=after_washing_hand.pct_deaths)

print(f'p-value is {p_value:.10f}')
print(f't-statstic is {t_stat:.4}')
```

Values:

```
p-value is 0.0000002985
t-statstic is 5.512
```

The calculated p-value (0.0000002985) is extremely small and well below the 1% threshold which suggests that the probability of observing such a large difference by chance is practically zero.

This provides highly strong compelling statistical evidence that the decline in mortality rates following the introduction of antisepsis actually occurred and was not due to random fluctuations.

Taken together with the high t-value (5.512), the results strongly support the conclusion that Semmelweis's hand-washing protocol led to a profound and statistically significant improvement in patient survival rates.

The Tragic Story of Dr Semmelweis

Despite the stunning evidence supporting Dr. Semmelweis's theory—that childbed (puerperal) fever was caused by a “substance” (which we now know to be bacteria) from corpses in the morgue—it was rejected by the medical community of the time.

One reason for this rejection was that Semmelweis wasn't particularly tactful. He gave the impression that doctors were transmitting childbed fever to women (which they actually were). That wasn't something people wanted to hear.

However, he also published his data in the form of long tables without any data visualizations:

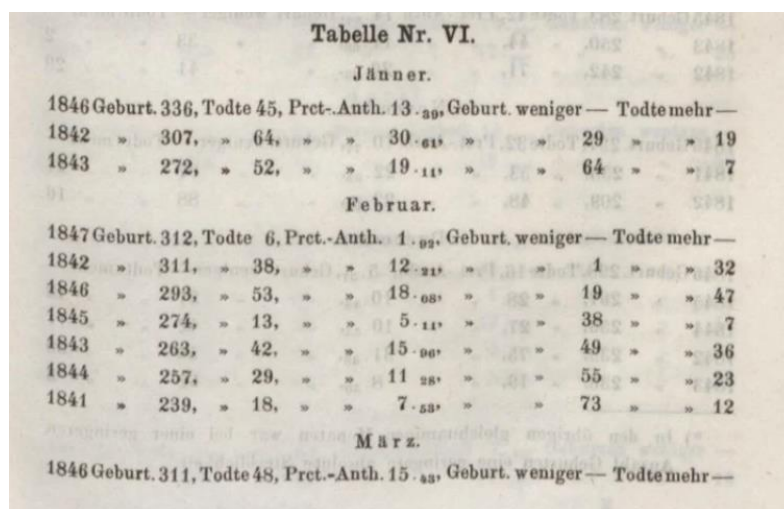


Tabelle Nr. VI.

Jänner.

1846	Geburt.	336,	Todte	45,	Prct.-Anth.	13.39,	Geburt. weniger —	Todte mehr —
1842	"	307,	"	64,	"	30.61,	"	29
1843	"	272,	"	52,	"	19.11,	"	64

Februar.

1847	Geburt.	312,	Todte	6,	Prct.-Anth.	1.92,	Geburt. weniger —	Todte mehr —
1842	"	311,	"	38,	"	12.21,	"	1
1846	"	293,	"	53,	"	18.08,	"	19
1845	"	274,	"	13,	"	5.11,	"	38
1843	"	263,	"	42,	"	15.96,	"	49
1844	"	257,	"	29,	"	11.28,	"	55
1841	"	239,	"	18,	"	7.53,	"	73

März.

1846	Geburt.	311,	Todte	48,	Prct.-Anth.	15.43,	Geburt. weniger —	Todte mehr —
------	---------	------	-------	-----	-------------	--------	-------------------	--------------

Because of the long tables, it was very hard to tell what was actually going on! Besides, statistics and statistical arguments were still quite uncommon in medicine back then.

Ultimately, Dr. Semmelweis's aggressive campaign earned him some powerful and influential enemies. He lost his position at the Vienna hospital, and the doctors stopped washing their hands with chlorine. The older Dr. Semmelweis got, the angrier he became, and he eventually began to behave rather “strangely.” This was either due to immense frustration or possibly the result of another illness, such as Alzheimer's or syphilis. In 1865, at the age of 47, Dr. Semmelweis was admitted to a psychiatric hospital. And he was likely beaten at the asylum, as he ultimately died of sepsis, a complication of an infection in the bloodstream. The tragic irony is that sepsis is a condition similar to the one he had fought so hard to prevent in women who were dying of childbed fever. It wasn't until 20 years later, thanks to Louis Pasteur's work on germ theory, that Dr. Semmelweis's findings gained recognition.

Rest in peace, Dr. Semmelweis, aka Savior of Mothers.

Conclusion

This project examined the profound impact of the antiseptic measures introduced by Ignaz Semmelweis. Historical hospital data were used to analyze trends in maternal mortality before and after the introduction of handwashing with a chlorine solution. By combining statistical analysis with data visualization, we were able to identify a clear and significant decline in mortality rates, demonstrating the crucial importance of hygiene in medical practice even before the scientific understanding of germs had been established.

From a data science perspective, this analysis followed a comprehensive and structured workflow that began with data preparation and transformation using DataFrames, followed by feature engineering, such as the calculation of mortality rates and the segmentation of the data based on a key intervention point. We applied various analytical methods, including time-series analysis, moving averages, distribution comparisons (box plots, histograms, KDE), and hypothesis tests using a t-test to assess statistical significance. These methods enabled us to derive meaningful insights from raw data by combining exploratory data analysis with formal statistical inference. Overall, the project demonstrates that it is possible to clean, analyze, visualize, and interpret real-world data, drawing well-founded conclusions from a variety of complementary approaches.

References

<https://www.udemy.com/course/100-days-of-code/>

https://pandas.pydata.org/docs/user_guide/index.html#user-guide

<https://matplotlib.org/stable/users/index.html>

<https://plotly.com/python/getting-started/>

<https://plotly.com/python/line-and-scatter/>

<https://plotly.com/python/histograms/>

<https://plotly.com/python/distplot/>

https://de.wikipedia.org/wiki/Ignaz_Semmelweis

<https://www.npr.org/sections/health-shots/2015/01/12/375663920/>